



Summary of the 2022 Clear Lake HAB Events

The Clear Lake Cyanotoxin Monitoring Program <https://bit.ly/ClearLakeCyanoProgram> which was begun by Big Valley and Elem Indian Colony in 2014 continued with its biweekly sampling for summer 2022. Despite the drought and low lake levels, we did not see the extremely elevated results for microcystin cyanotoxin as we did during the 2021 sampling events. In fact, the highest toxin level on the lake was 790 µg/L and was in the Oaks arm of the Lake.

The neurotoxin Anatoxin-a was noted in one site during the May – November sampling period. Saxitoxin, also a neurotoxin, was noted at low levels in all three arms during September – November. Because saxitoxin has not been present during previous years, the Tribe is doing further review of the cyanobacteria genera and water quality conditions in those areas with saxitoxin presence.

The cyanotoxin blooms varied throughout the year, with *Microcystis sp.* dominating the majority for most of the year, but with regular occurrences of *Lyngbya sp.*, *Planktothrix sp.* and *Gloeotrichia sp.*



2022 Clear Lake Cyanotoxin Sampling Signage Recommendations (May – November)

SITE ID	5/10 F	6/1 F	6/14 F	7/11/F	7/26 F	8/9 F	8/23 F	9/7 F	9/20 F	10/04 F	10/18 F	11/01 F	11/15 F
AP01	DANGER	NONE	WARNING	DANGER	DANGER	DANGER	DANGER	CAUTION	WARNING	WARNING	NONE	CAUTION	n/a
BP	NONE	NONE	NONE	NONE	DANGER	DANGER	CAUTION	WARNING	CAUTION	NONE	NONE	NONE	NONE
BVCL6	NONE	NONE	NONE	NONE	NONE	NONE	CAUTION	NONE	CAUTION	NONE	NONE	NONE	NONE
CL-1	NONE	N/A	NONE	NONE	N/A	NONE	NONE	NONE	N/A	NONE	NONE	NONE	n/a
CL-3	NONE	N/A	CAUTION	CAUTION	N/A	CAUTION	CAUTION	NONE	N/A	CAUTION	CAUTION	NONE	n/a
CL-4	NONE	N/A	CAUTION	CAUTION	N/A	CAUTION	CAUTION	CAUTION	N/A	DANGER	DANGER	NONE	n/a
CLOAKS01	CAUTION	NONE	WARNING	CAUTION	DANGER	WARNING	CAUTION	CAUTION	DANGER	DANGER	CAUTION	CAUTION	NONE
CLV7	NONE	NONE	NONE	CAUTION	WARNING	DANGER	DANGER	DANGER	CAUTION	NONE	NONE	NONE	n/a
CP	NONE	NONE	NONE	NONE	CAUTION	CAUTION	CAUTION	NONE	NONE	NONE	NONE	NONE	n/a
ELEM01	CAUTION	CAUTION	DANGER	CAUTION	DANGER	DANGER	CAUTION	NONE	WARNING	WARNING	NONE	CAUTION	NONE
GH	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	DANGER	NONE	NONE	NONE	n/a
HB	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	CAUTION	NONE	NONE	NONE	n/a
JB	DANGER	DANGER	WARNING	DANGER	CAUTION	WARNING	CAUTION	CAUTION	NONE	CAUTION	CAUTION	NONE	NONE
KEYS03	NONE	NONE	NONE	NONE	NONE	N/A	N/A	N/A	N/A	n/a	n/a	N/A	n/a
KP01	NONE	NONE	NONE	NONE	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	NONE	NONE
LA-03	N/A	NONE	CAUTION	N/A	N/A	CAUTION	N/A	N/A	CAUTION	NONE	n/a	CAUTION	NONE
LC01	N/A	NONE	NONE	NONE	WARNING	NONE	CAUTION	CAUTION	CAUTION	CAUTION	NONE	NONE	CAUTION
LPTNT	NONE	NONE	NONE	NONE	CAUTION	CAUTION	CAUTION	CAUTION	WARNING	WARNING	CAUTION	NONE	CAUTION
LUC01	NONE	DANGER	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	n/a
NR-02	N/A	NONE	NONE	N/A	N/A	CAUTION	N/A	N/A	CAUTION	NONE	n/a	CAUTION	NONE
OA-04	N/A	NONE	CAUTION	N/A	N/A	WARNING	N/A	N/A	WARNING	CAUTION	CAUTION	NONE	CAUTION
RED01	WARNING	CAUTION	DANGER	CAUTION	WARNING	DANGER	DANGER	WARNING	DANGER	DANGER	DANGER	DANGER	WARNING
RODS	NONE	NONE	NONE	NONE	NONE	CAUTION	NONE	NONE	N/A	n/a	n/a	N/A	n/a
SBMMEL01	DANGER	CAUTION	DANGER	WARNING	DANGER	DANGER	WARNING	NONE	CAUTION	CAUTION	NONE	NONE	n/a
SHADY01	NONE	NONE	DANGER	CAUTION	WARNING	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	CAUTION	NONE	CAUTION
UA-01	N/A	NONE	NONE	N/A	N/A	CAUTION	N/A	N/A	CAUTION	NONE	n/a	CAUTION	NONE
UA-06	N/A	NONE	NONE	N/A	N/A	CAUTION	N/A	N/A	CAUTION	NONE	n/a	CAUTION	NONE
UA-08	N/A	NONE	NONE	N/A	N/A	NONE	N/A	N/A	NONE	NONE	n/a	NONE	NONE

All cyanotoxin monitoring conducted by Big Valley Band of Pomo Indians. Signage recommendations are based on lab results of microcystin toxin values at the site. https://mywaterquality.ca.gov/habs/resources/habs_response.html#advisory_signs_guidance

N/A: not sampled None: < 0.8 µg/l Caution: 0.8 ug/L to <6.0 µg/l Warning: 6.0 µg/l to <20 µg/l Danger: > 20 µg/l



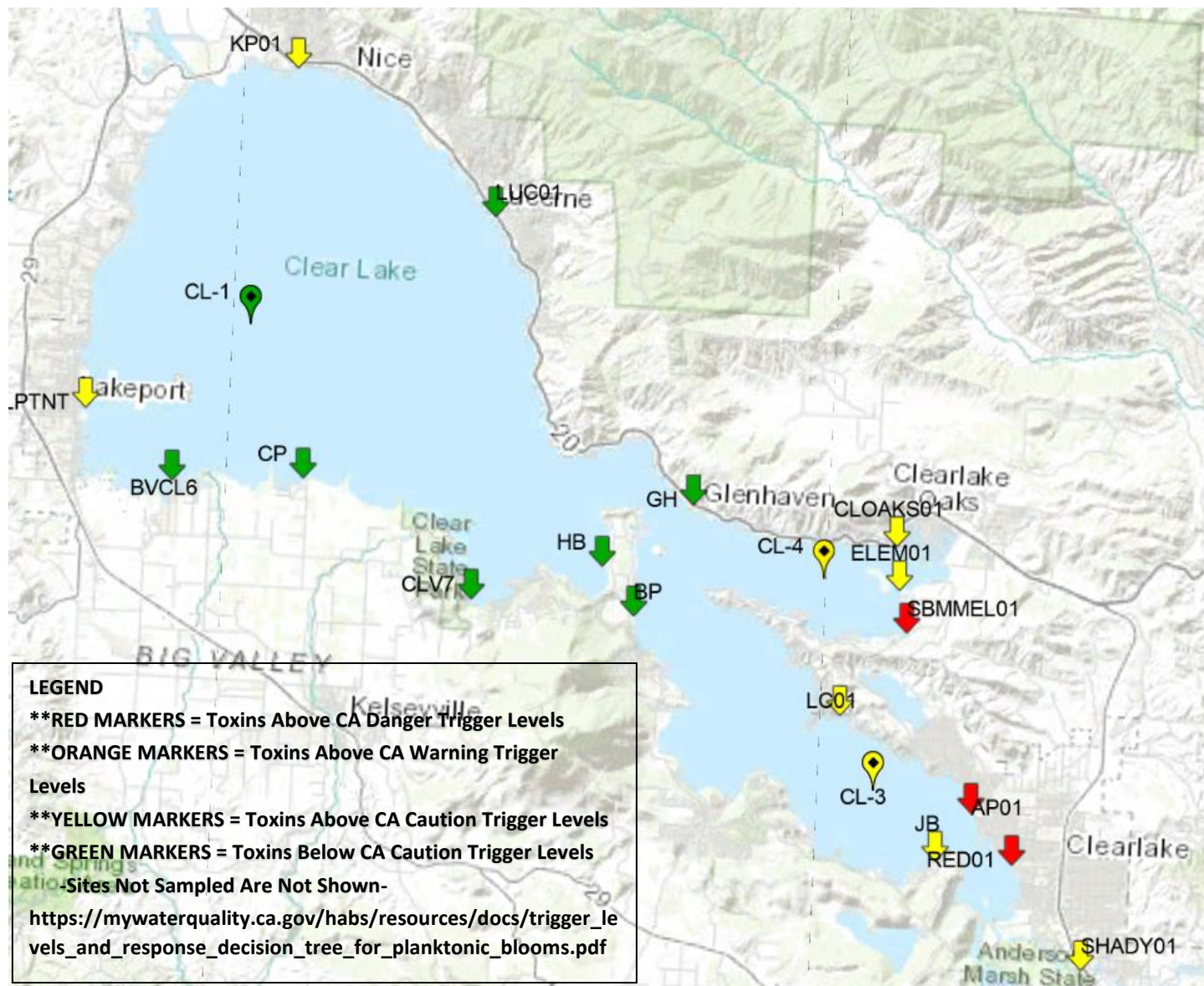
Microcystin and Anatoxin Results for May – November 2022 Clear Lake Sampling

		MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA	MC	ANA
SITE ID	ARM	5/10	5/10	6/1/	6/1/	6/14	6/14	7/11	7/11	7/26	7/26	8/9	8/9	8/23	8/23	9/7	9/7	9/20	9/20	10/4	10/4	10/18	10/18	11/1	11/1	11/15	11/15
AP01	L	0.21	N/A	0.80	ND	6.14	N/A	147.81	N/A	176	N/A	489.28	N/A	62.67	ND	0.82	N/A	8.07	N/A	7.70	n/a	0.57	ND	1.30	ND	n/a	n/a
BP	L	ND	N/A	N/A	N/A	N/A	N/A	N/A	N/A	45.53	N/A	268.63	N/A	1.65	ND	18.95	N/A	5.99	N/A	0.52	n/a	0.25	ND	0.40	ND	0.34	ND
BVCL6	U	N/A	N/A	N/A	N/A	N/A	N/A	0.39	ND	0.66	N/A	0.63	N/A	0.87	ND	0.69	N/A	1.48	ND	0.23	n/a	0.20	ND	0.28	ND	0.22	ND
CL-1	U	N/A	N/A	N/A	N/A	0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.15	n/a	n/a	n/a	ND	ND	n/a	n/a
CL-3	L	N/A	N/A	N/A	N/A	3.45	N/A	3.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.78	n/a	n/a	n/a	0.26	ND	n/a	n/a
CL-4	O	N/A	N/A	N/A	N/A	2.11	N/A	2.62	N/A	N/A	N/A	2.20	N/A	N/A	N/A	1.33	N/A	N/A	N/A	35.46	n/a	n/a	n/a	0.41	ND	n/a	n/a
CLOAKS01	O	2.80	N/A	0.15	N/A	16.87	N/A	2.97	N/A	242.40	N/A	18.25	N/A	3.95	ND	0.88	N/A	68.05	N/A	25.17	n/a	0.80	ND	2.22	ND	0.40	ND
CLV7	U	N/A	N/A	N/A	N/A	N/A	N/A	1.73	N/A	8.13	N/A	33.19	N/A	70.82	ND	51.80	N/A	1.87	N/A	0.23	n/a	ND	ND	0.22	ND	n/a	n/a
CP	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.22	N/A	2.08	N/A	1.02	ND	0.32	N/A	0.21	N/A	0.23	n/a	ND	ND	ND	ND	n/a	n/a
ELEM01	O	5.16	N/A	1.56	N/A	84.21	N/A	1.72	N/A	636	N/A	223.89	N/A	1.37	ND	N/A	N/A	6.69	N/A	18.18	n/a	0.18	ND	0.97	ND	0.52	ND
GH	O	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	64.17	N/A	0.44	n/a	0.23	ND	0.40	ND	n/a	n/a
HB	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.39	n/a	ND	ND	0.22	ND	n/a	n/a	
JB	L	0.51	N/A	45.91	ND	17.62	N/A	27.41	N/A	3.88	N/A	13.97	N/A	5.26	ND	1.00	N/A	0.45	N/A	5.21	n/a	1.16	ND	0.52	ND	0.55	ND
KEYS03	O	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	n/a	n/a	n/a	n/a	N/A	N/A	n/a	n/a
KP01	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.35	N/A	1.90	N/A	1.26	ND	3.38	N/A	2.27	N/A	1.30	n/a	2.89	ND	0.24	ND	0.19	ND
LA-03	L	N/A	N/A	0.38	N/A	4.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.61	n/a	n/a	n/a	1.15	ND	0.76	ND
LC01	L	N/A	N/A	0.24	N/A	N/A	N/A	N/A	N/A	6.30	N/A	0.49	N/A	5.73	ND	3.60	N/A	2.18	N/A	2.49	n/a	0.78	ND	0.36	ND	1.17	ND
LPTNT	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.83	N/A	2.25	N/A	4.22	ND	3.39	N/A	19.10	N/A	6.15	n/a	0.94	ND	0.17	ND	2.96	ND
LS2	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	n/a	n/a	n/a	n/a	N/A	N/A	n/a	n/a
LUC01	U	N/A	N/A	22.66	N/A	0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.31	ND	N/A	N/A	N/A	N/A	0.26	n/a	0.24	ND	ND	ND	n/a	n/a
NR-02	O	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.35	n/a	n/a	n/a	N/A	N/A	0.44	ND
OA-04	O	N/A	N/A	N/A	N/A	2.69	N/A	N/A	N/A	N/A	N/A	6.01	N/A	N/A	N/A	N/A	N/A	9.08	ND	1.19	n/a	n/a	n/a	0.59	ND	0.95	ND
RED01	L	0.17	N/A	1.37	ND	39.40	N/A	0.90	N/A	11.79	N/A	113.70	N/A	101.25	ND	14.59	N/A	84.01	N/A	31.60	n/a	32.28	ND	39.46	ND	6.29	ND
RODS	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.81	N/A	0.46	ND	N/A	N/A	N/A	N/A	n/a	n/a	n/a	n/a	N/A	N/A	n/a	n/a
SBMMEL01	O	34.58	0.20	4.91	ND	790.00	N/A	11.27	N/A	118	N/A	64.06	N/A	11.50	ND	0.73	N/A	1.98	N/A	3.32	n/a	0.43	ND	0.54	ND	n/a	n/a
SHADY01	L	0.63	N/A	N/A	N/A	22.81	N/A	1.58	N/A	13.39	N/A	3.10	N/A	1.50	0.38	1.24	ND	1.01	N/A	0.83	n/a	1.74	ND	0.68	ND	5.01	ND
UA-01	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.25	n/a	n/a	n/a	N/A	N/A	0.19	ND
UA-06	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.20	n/a	n/a	n/a	N/A	N/A	0.21	ND
UA-08	U	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.42	n/a	n/a	n/a	0.16	ND	0.16	ND
UBL		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	n/a	n/a	n/a	n/a	N/A	N/A	n/a	n/a
# sites with highest results that sampling event (counted when results come in)		5	1	4	0	11	0	4	0	10	0	8	0	3	1	1	0	3	0	5	0	0	0	0	0	1	0

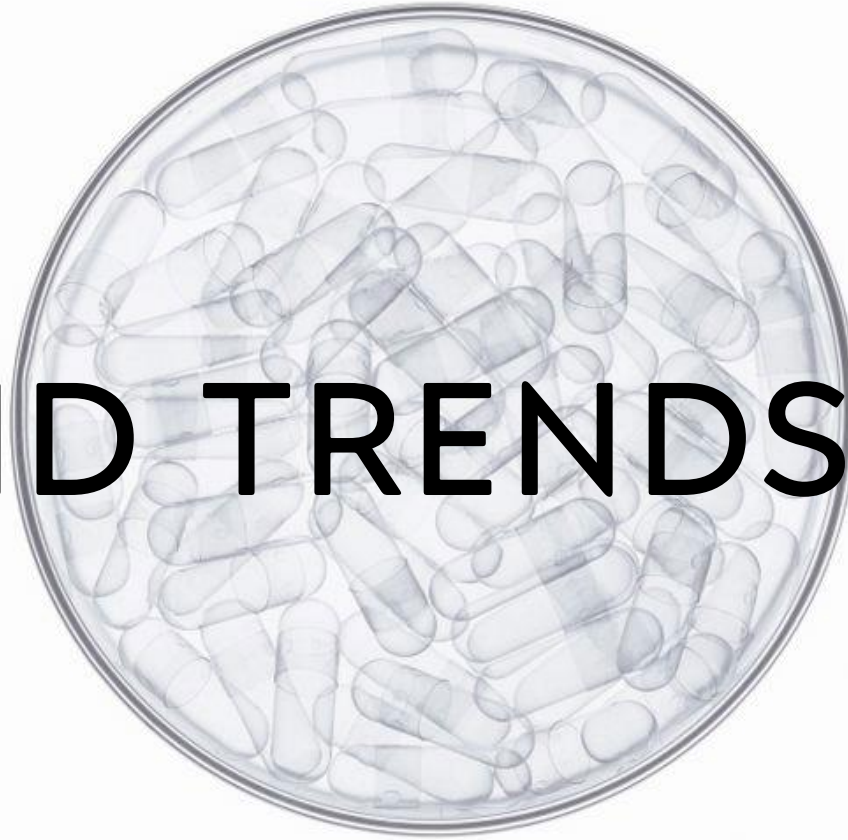
This table shows the results for both microcystin (MC) and Anatoxin-a (ANA) analysis at each site. Cells that are highlighted brown have the highest toxin value that sampling date. Cells that are dotted are the highest toxin value for that site during the 2022 sampling year.



Clear Lake Summer 2022 (May – October)
Site Specific Most Frequent Toxin Level
(for regularly sampled sites)



CELL ID TRENDS: 2022



BY: BIG VALLEY RANCHERIA

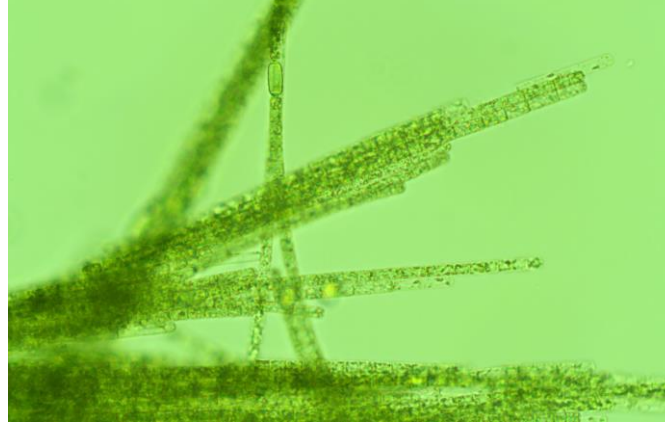
MAY, 2022

BIG VALLEY SAMPLED 17-19 SITES ON THE SHORELINE OF CLEAR LAKE ON A MONTHLY BASIS FOR MAY-OCTOBER

MICROSCOPY: 5/10 & 5/30

- 5/10: DOMINANT: APHANIZOMENON
- 5/10: CO-DOMINANT: MICROCYSTIN
- 5/30: DOMINANT: WORONICHINA
- 5/30: CO-DOMINANT: DOLICHOSPERMUM

Aphanizomenon - CP 5/10/2022



Woronichinia - SHADY01 5/30/2022



Microcystis - ELEM01 5/30/2022



Dolichospermum - CLOAKS01 5/10/2022

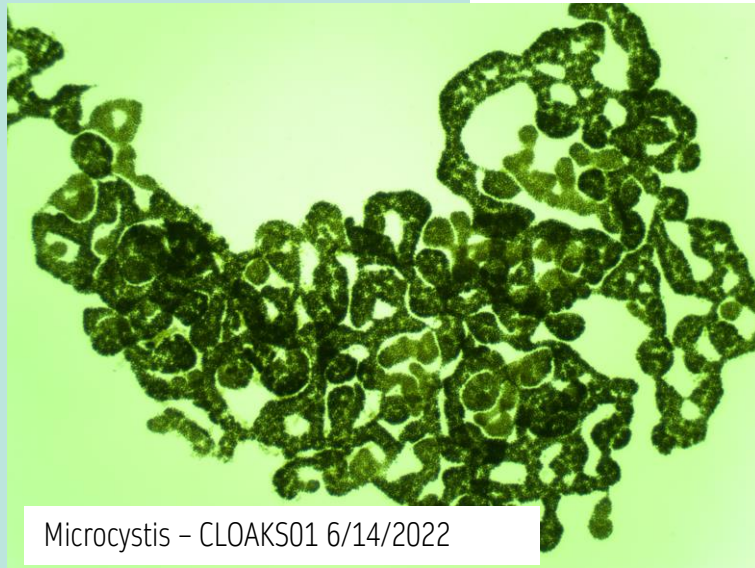


JUNE, 2022

BIG VALLEY SAMPLED 17-19 SITES ON THE SHORELINE OF
CLEAR LAKE ON A MONTHLY BASIS FOR MAY-OCTOBER

MICROSCOPY: 6/14

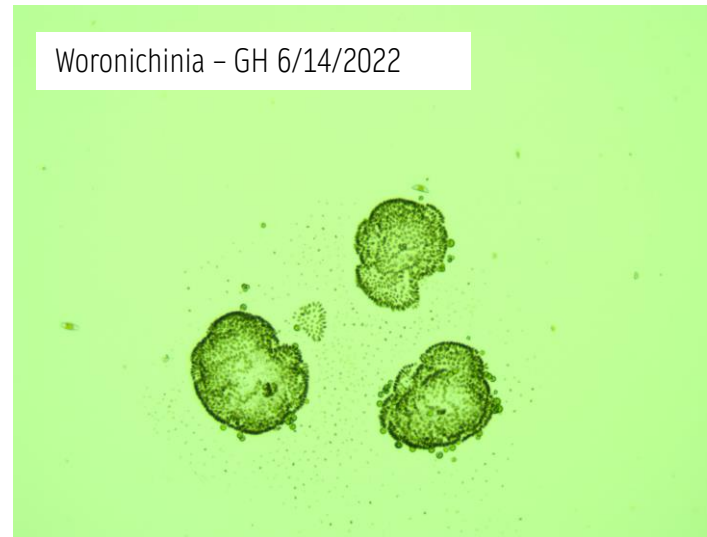
- DOMINANT: MICROCYSTIN
- CO-DOMINANT: WORONICHINIA & APHANIZOMENON



Microcystis - CLOAKS01 6/14/2022



Aphanizomenon - LA03 6/14/2022



Woronichinia - GH 6/14/2022

JULY, 2022

BIG VALLEY SAMPLED 17-19 SITES ON THE SHORELINE OF CLEAR LAKE ON A MONTHLY BASIS FOR MAY-OCTOBER

MICROSCOPY: 7/11 & 7/26

- 7/11: DOMINANT: MICROCYSTIN
- 7/11: CO-DOMINANT: GLOEOTRICHIA
- 7/26: DOMINANT: LYNGBYA
- 7/26: CO-DOMINANT: MICROCYSTIN AND GLOEOTRICHIA

Lyngbya – RED01 7/26/2022



Gloeotrichia – BP 7/11/2022



Microcystis – ELEM01 7/11/2022



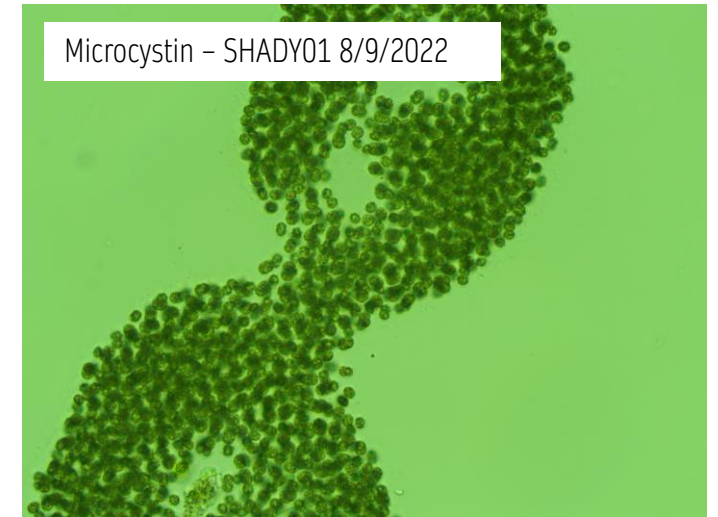
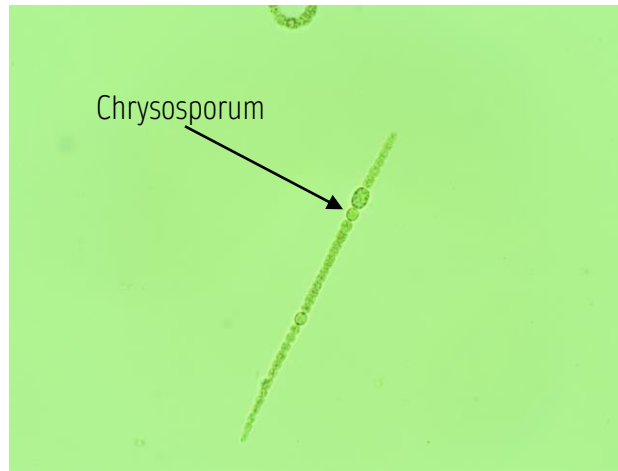
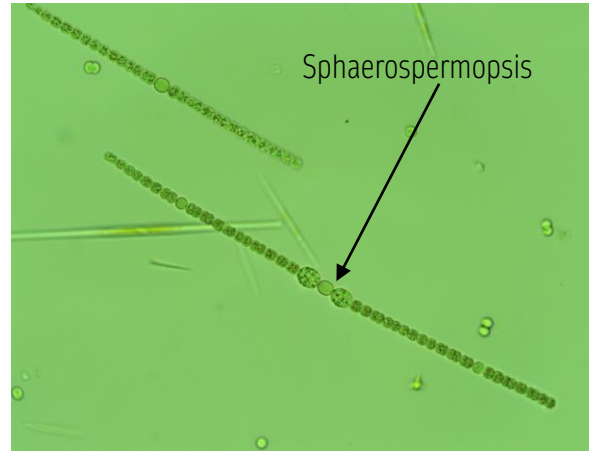
AUGUST, 2022

Big Valley sampled 17-19 sites on the shoreline of clear lake
on a monthly basis for May-October

Microscopy: 8/9

- Dominant: Microcystin
- Co-dominant: Lyngbya

Big Valley also began noticing a couple of other cyanobacteria, Chrysosporum and Sphaerospermopsis at most all sites.

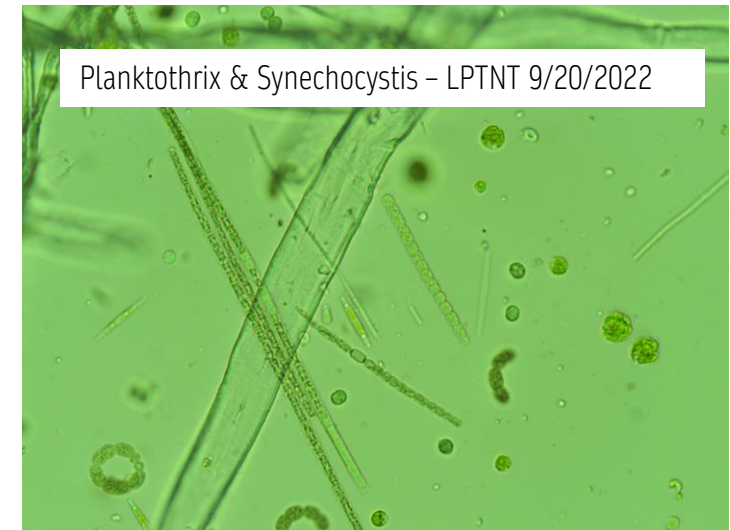


SEPTEMBER, 2022

Big Valley sampled 17-19 sites on the shoreline of Clear Lake on a monthly basis for May-October

Microscopy: 9/7 & 9/20

- Dominant: Microcystin
- Co-dominant: Anabaenopsis, Synechocystis, and Planktothrix



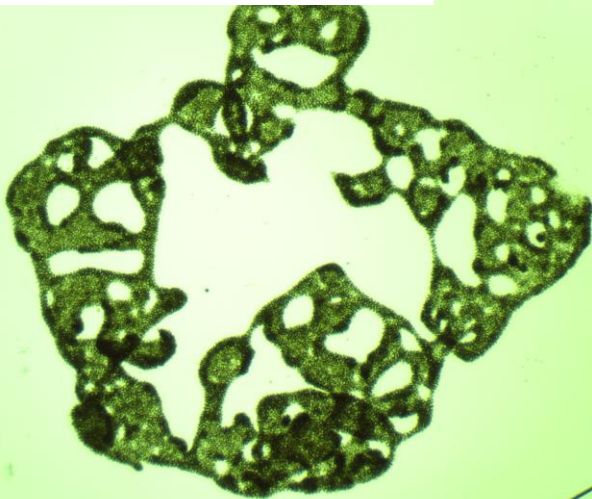
OCTOBER, 2022

Big Valley sampled 17-19 sites on the shoreline of clear lake on a monthly basis for May-October

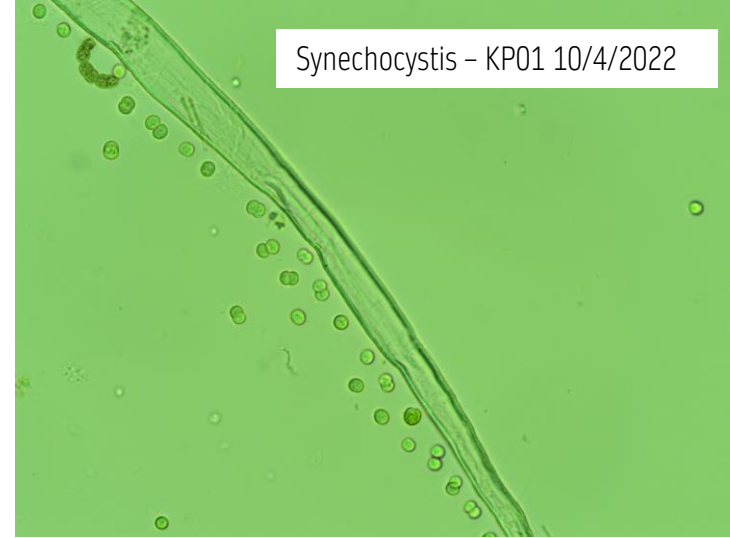
Microscopy: 10/4 & 10/18

- Dominant: Microcystin
- Co-dominant: Synechocystis & Planktothrix

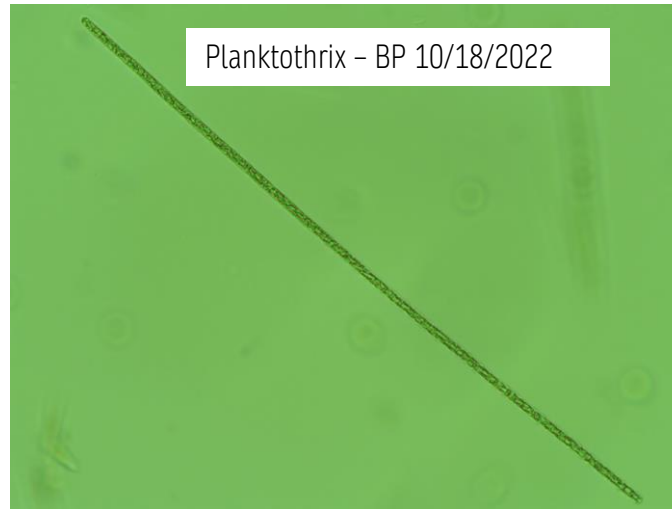
Microcystis – SBMMEL01 10/4/2022



Synechocystis – KP01 10/4/2022



Planktothrix – BP 10/18/2022



Planktothrix & Synechocystis – CLV7 10/18/2022

